Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical AI

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***Abstract*—Hospital FL faces gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under multi-year retention. ZKFL-PQ combines NIST ML-KEM-768 transport; an Unruh NIZK (default r=128) of ℓ_2-bounded updates in language $\mathcal{L}_\tau^{\text{bind}}$ with an Enc-consistency gadget for BFV coins; full-vector BFV aggregation opened by (t, n)-threshold *partial* decryption (sk never reconstructed); and a default post-ZKP coordinate-wise median against sign-flip and in-bound backdoors that pure ℓ_2 gates must accept. A public artifact provides fused SEAL+threshold HE, Lean/Python/EasyCrypt combinatorial Unruh checks, and communication accounting under varying Unruh repetitions and HE degrees. Empirically we report multi-seed synthetic baselines, scale N=20/T=30, UCI Breast Cancer, full-resolution PneumoniaMNIST, and backdoor attack success rates for ZKP+median vs. FedAvg/ℓ_2-only. ZKFL-PQ is complementary to Beskar-class secure aggregation rather than a replacement.**

***Index Terms*—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness**

I. Introduction

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that ask how updates are authenticated and who can decrypt them. Wrapping FedAvg in classical TLS is insufficient against adversaries that store ciphertexts today and open them later [4].

a) Contributions.:

1) A layered protocol that binds Unruh norm proofs to BFV ciphertexts via $\mathcal{L}_\tau^{\text{bind}}$ + Enc-consistency, opens with threshold partial decrypt, then robust-aggregates (default median), with each layer addressing a distinct threat (Table I).
2) Concrete Unruh (r=128) and BFV parameters, a named soundness/privacy theorem, and Lean/Python/`QROM.ec` combinatorial checks, plus measured payloads under varying r and HE degree.
3) A reproducible artifact [12]: full-vector encrypt+prove, crypto-only accept, fused SEAL+threshold path, no monolithic server sk.
4) An empirical study spanning large-norm and sign-flip attacks, backdoor ASR with ZKP+median (Fig. 3), UCI Breast Cancer, full-resolution MedMNIST, and scale N=20, T=30.

TABLE I
Threat coverage by layer (composition is the contribution).

Threat	ML-KEM	ZKP ℓ_2	Enc-cons.	Thr. open	Median
HNDL / PQ transit	yes	–	–	–	–
Oversized update	–	yes	–	–	–
Proof/ct split	–	bind	yes	–	–
Curious aggregator	–	–	–	yes	–
Sign-flip / backdoor	–	no‡	–	–	yes

‡In-bound updates satisfy τ; Table IV reports 0% crypto detection.

II. Threat Model and Claims

Adversary: (i) network eavesdropper with future quantum computers against factoring/DLP; (ii) up to $f<N/2$ Byzantine clients; (iii) curious aggregator that sees protocol messages but does not hold sk alone; (iv) up to $t-1$ colluding threshold decryptors.

Claims: ML-KEM-768 transport (FIPS 203 Level 3) [5]; norm soundness for oversized witnesses under SIS + (Unruh/)RO assumptions; ciphertext binding so a proof cannot be replayed on a different **ct**; aggregator privacy via Shamir-shared sk and partial decryption $\mu_i = c_1 \star (\lambda_i \cdot s_i)$.

Scope. We do not claim differential privacy or a drop-in replacement for Beskar-class full-stack secure aggregation [8]. ZKFL-PQ targets verifiable norm gating, multi-party decrypt, and robust post-filtering; pure ℓ_2 alone does not stop sign-flip or in-bound backdoors (Table I).

III. Related Work

Beskar offers a mature PQ secure-aggregation stack with assisting nodes and DP [8]. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets

TABLE II
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

System	PQ transport	Norm ZKP	HE / thr. dec.
Beskar [8]	yes (masking/DP)	–	secure agg. stack
Patel [9]	lattice HE	–	HE FL
Lu et al. [10]	hybrid PQ	–	HE FL
Krum [3]	–	statistical	–
zkML [11]	–	inference ZK	–
ZKFL-PQ (ours)	ML-KEM-768	Unruh+bind	BFV(t, n)

inference proofs rather than FL aggregation [11]. Multi-Krum remains effective against low-norm adversaries; ZKFL-PQ therefore composes cryptographic accept with median/Krum rather than relying on ℓ_2 alone.

IV. Protocol Design

A. Round Structure

Each round t: the server broadcasts $w^{(t)}$. Client i runs local SGD on D_i, quantizes to $\tilde{w}_i$, encrypts under BFV public key pk, proves $\|\tilde{w}_i\|_2 \leq \tau$ bound to $\mathbf{ct}_i$ (and Enc-consistency of coins), and sends an ML-KEM-wrapped payload. The server verifies proofs, threshold-opens each accepted ciphertext, and applies coordinate-wise median (default) before the model step (Algorithm 1).

B. Quantization

Quantize clips coordinates to $[-c, c]$, scales by s, and rounds into $\mathbb{Z}_t$ (artifact defaults c=1, s=10^3). The public threshold τ applies after quantization; adaptive rules $\tau^{(t)} = \hat{\mu}+k\hat{\sigma}$ are compatible. Reported runs use fixed τ (synthetic 5; medical 8–12).

C. Bound Norm Language

Definition 1 (Bound language)**.** *Let* $C = \mathsf{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w}\|\mathbf{r}) \bmod q'$ *and* $\mathbf{ct} = \mathsf{Enc}_{pk}(\tilde{w}; \rho)$*. Then*

$$\mathcal{L}_\tau^{\text{bind}} = \big\{(C, \mathbf{ct}, \tau) : \exists(\tilde{w}, \mathbf{r}, \rho) \\ C = \mathsf{Commit}(\tilde{w}; \mathbf{r}),\ \mathbf{ct} = \mathsf{Enc}(\tilde{w}; \rho),\ \|\tilde{w}\|_2 \leq \tau\big\}. \quad (1)$$

a) Σ-protocol [6].: Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C\|\mathbf{ct}\|T\|\tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy $T + eC \equiv \mathbf{A}(\mathbf{z}\|\mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \leq B$. Rejection sampling yields HVZK in the classical ROM.

b) Unruh-oriented NIZK.: Classical FS is not tightly QROM-secure [7]. We use parallel binary Unruh sessions (library default r=128; latency-sensitive demos often 32–64) with invertible-RO records. Combinatorial uniqueness / 2^{-r} lemmas are machine-checked in Lean 4; executable game hops and a SHA3-instantiated EasyCrypt QROM library (`formal/easycrypt/lib/{SHA3,QROMCore}.ec`) accompany the artifact.

c) Enc-consistency.: Beyond hashing $\mathbf{ct}$ into the challenge, a dedicated Σ-gadget proves knowledge of coins $\rho = (u, e_0, e_1)$ such that the BFV encryption equations hold for the same plaintext chunk, closing split attacks that pair a benign proof with a mismatched ciphertext.

TABLE III
CRYPTOGRAPHIC PARAMETERS (RESEARCH STACK).

Component	Setting	Notes
ML-KEM-768	n=256, k=3, q=3329	FIPS 203 L3 [5]
BFV degree	n=512 / 4096	Lightweight vs. `classic128`
BFV moduli	q≈2^{40}, t=2^{12}	NumPy encoder; optional estimator / SEAL
Threshold	(t, n)=$(2, 3)$ partial decrypt	Server sk=$\perp$
Unruh reps	r=128 default	Latency-sensitive runs may use r=64
Post-ZKP agg	median (default)	Multi-Krum / mean overrides
Packing	full-vector chunks	All d coordinates encrypted

Algorithm 1 ZKFL-PQ round (default composition)

Require: $w^{(t)}$, τ, KEM (ek_S, dk_S), HE pk, decryptor shares
Broadcast $w^{(t)}$
for each client i **do**
  $\tilde{w} \leftarrow$ Quantize(LocalSGD($w^{(t)}, D_i$))
  $\mathbf{ct}_i \leftarrow \mathsf{Enc}_{pk}(\tilde{w}; \rho_i)$; prove Enc-consistency of ρ_i
  $\pi_i \leftarrow \mathsf{Prove}_{\mathsf{Unruh}}((\mathbf{ct}_i, \tau), \tilde{w})$
  send ML-KEM-wrapped $(\mathbf{ct}_i, \pi_i)$
end for
Keep i iff cryptographic verify returns 1
For each accepted i: threshold-open $\mathbf{ct}_i$ to recover Δ_i
$\bar{\Delta} \leftarrow$ Median($\{\Delta_i\}$) ▷ or Multi-Krum; not plaintext mean alone
$w^{(t+1)} \leftarrow w^{(t)} + \bar{\Delta}$

D. BFV Aggregation and Threshold Opening

Additive BFV yields $\mathsf{Dec}(\sum_i \mathsf{Enc}(m_i)) = \sum_i m_i$ under a noise budget growing roughly as $N \cdot O(\sigma\sqrt{n})$. After key generation, secret coefficients are Shamir-shared and the monolithic sk is erased. Opening uses Lagrange-weighted partial decryptions with smudging: s is never reconstructed for production decrypt. An optional TenSEAL/SEAL sidecar runs in the same process as threshold BFV (`FusedSealThresholdHE`): primary open remains partial-decrypt, while SEAL provides a certified encoder path and consistency check.

Theorem 1 (Composition soundness and threshold privacy)**.** *Assume SIS binding of* Commit*, RLWE semantic security of BFV, and RO modeling of* H=SHA3-256*. Then: (i) honest* $\|\tilde{w}\|_2 \leq \tau$ *proofs are accepted except for rejection-sampling aborts; (ii) an oversized witness is accepted with probability at most* $\approx 2^{-r}$ *(binary Unruh) plus SIS advantage; (iii) Enc-consistency plus digesting* $\mathbf{ct}$ *into challenges prevent attaching a valid proof to a mismatched ciphertext; (iv) the aggregator and any* $t-1$ *decryptors learn nothing beyond what* t *partial opens reveal. Combinatorial Unruh uniqueness/*2^{-r} *is machine-checked in Lean 4; EasyCrypt theories instantiate* H *as SHA3-256 with an O2H-style* $q(q+1)/2^{256}$ *term (`formal/run_formal_ci.py`).*

a) Implementation.: The public repository [12] realizes Algorithm 1: shared prove/encrypt dimension, ciphertext-

TABLE IV

MEAN±STD OVER 5 SEEDS (SYNTHETIC). ACCURACY (%) AND DETECTION.

Method	Acc. (large)	Acc. (flip)	Det. (large)	Det. (flip)
FedAvg	26.0±1.0	41.0±13.2	0%	0%
Clip@τ	100±0	40.3±12.7	–	–
Multi-Krum	100±0	100±0	–	–
HE-only	22.1±2.1	19.8±9.5	0%	0%
ZKP-only	84.9±30.2	40.9±13.2	97%	0%‡
Hybrid+mean	87.0±26.0	40.1±12.6	97%	0%‡
Hybrid+median	100±0	100±0	97%	0%‡

‡Sign-flip respects τ—ℓ_2 proofs must accept (0% crypto detection). Hybrid+mean keeps plaintext mean after accept; Hybrid+median is the default composition and restores accuracy [12].

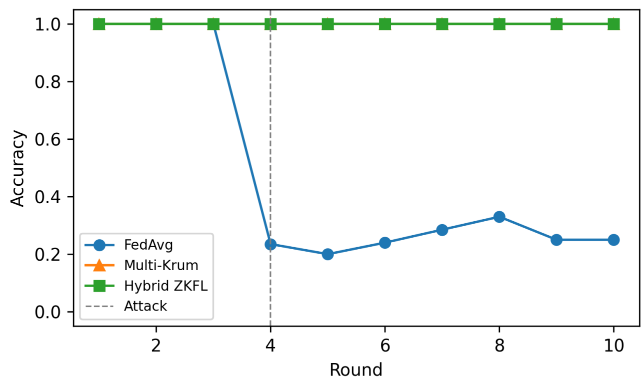

Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

bound challenges, crypto-only accept, full-vector packing, Enc-consistency, fused SEAL+threshold HE, and default median. Scripts pin seeds and emit JSON.

V. EVALUATION

A. Synthetic Multi-Baseline Study

Setup. Synthetic 4-class task (1,000 samples, 784 features, Dirichlet α=0.5), N=5, MLP with d=108,996, 10 rounds, seeds $\{42, \ldots, 46\}$. Attacks from round 4: (i) large-norm Gaussian; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$. Baselines: FedAvg, clip@τ, Multi-Krum (f=1), HE-only, ZKP-only, Hybrid+mean (privacy path without robust agg), and Hybrid+median (default composition).

Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table IV, Fig. 1–2). Sign-flip within τ is accepted by pure ℓ_2 and Hybrid+mean (0% crypto detection, Table I), while Multi-Krum and Hybrid+median recover accuracy by pairwise or coordinate-wise robust aggregation after cryptographic accept.

B. Scale, Medical, and Backdoor Studies

A compact-model scale study with N=20 clients and T=30 rounds reproduces the same qualitative pattern under both attacks: ZKP and hybrid reject oversized updates with high probability, while sign-flip still requires a robust aggregator.

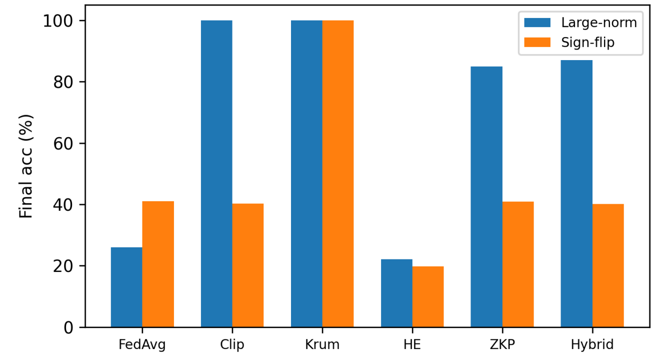

Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

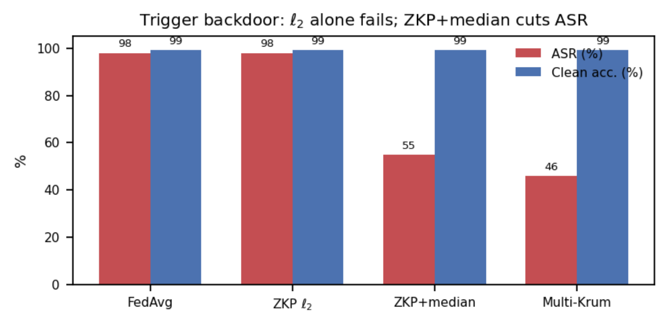

Fig. 3. Backdoor ASR vs. clean accuracy under FedAvg, ℓ_2-ZKP, ZKP+median, and Multi-Krum.

Wall-clock remains interactive on CPU for the compact MLP used in that study, which is the intended operating point for validating the crypto pipeline before scaling encoders.

On UCI Breast Cancer Wisconsin (569 samples, 30 clinical features; compact MLP with d=1554), the target stack—Unruh with r=64 (library default 128), full-vector fused HE, $(2, 3)$ partial threshold decrypt, and post-ZKP median—yields accuracies 88.6% $\rightarrow$ 93.0% $\rightarrow$ **93.9**% across three rounds while rejecting a scripted oversized client (Fig. 4). Mean round time is $\approx$ 4–9 s and per-round payload $\approx$ 1.4–1.7 MB on a laptop CPU. Full-resolution PneumoniaMNIST is evaluated with a compact MLP head ($\approx$12.7k parameters) and with `ConvNet28` ($\approx$51.6k parameters, no projection, no compact head); a smoke run of the CNN stack reaches $\approx$86.7% accuracy in one round while rejecting an oversized client (`run_medmnist_cnn.py`).

Trigger backdoors whose poisoned updates remain inside τ are accepted by pure ℓ_2 ZKPs (ASR $\approx$98%, matching FedAvg). Default ZKP+median cuts ASR to $\approx$55% while preserving clean accuracy $\approx$99%; Multi-Krum alone reaches $\approx$46% ASR (Fig. 3). Cryptographic accept filters oversized updates; statistical robust aggregation addresses in-bound poisons.

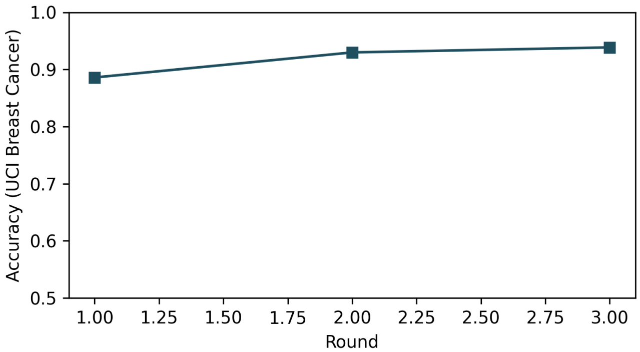

Fig. 4. UCI Breast Cancer accuracy under the full target stack.

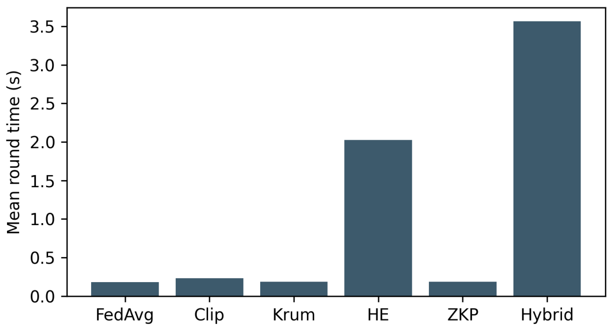

Fig. 5. Mean round time (synthetic hybrid $\sim$20$\times$ FedAvg).

C. Communication Cost

Full-vector HE dominates bandwidth. Uplink scales as $O(N\lceil d/n\rceil n \log q)$ bits plus Unruh $O(r \cdot (d + \ell))$. Table V: r=128 roughly doubles the proof component vs. r=64; `classic128` (n=4096) targets a Classic-128 HE parameter class at about 2$\times$ ciphertext volume for the UCI-sized model.

VI. Security Discussion

Honest clients with $\|\tilde{w}\|_2 \leq \tau$ produce Unruh sessions that pass algebraic and norm checks except for rejection-sampling aborts (bounded retries in the artifact). If $\|\tilde{w}\|_2 \gg \tau$, sessions with challenge bit 1 fail with overwhelming probability under the Gaussian mask design; soundness therefore depends on the fraction of 1-challenges across r repetitions. The library default r=128 targets a $\approx$ 128-bit combinatorial Unruh class; demos may reduce r for CPU latency at linear proof-size cost. Binding H to ciphertext bytes, together with the Enc-consistency gadget for ρ, prevents a benign proof from being attached to a different ciphertext. BFV semantic security (RLWE) hides individual updates from parties lacking sk; with Shamir (t, n), the aggregator and any $t-1$ decryptors cannot open. We do not claim differential privacy; sites that need DP can compose mechanisms as in Beskar [8].

TABLE V
Payload under Unruh/n settings (scaled accounting).

Setting	Msg/round	Notes
Synthetic hybrid	$\sim$100 KB	Reference
Medical r=64, n=512	1.4–1.7 MB	Lightweight HE
Medical r=128, n=512	$\approx$2.0–2.5 MB	Default Unruh
Medical r=128, n=4096	$\approx$3–4 MB	Classic-128 class
Plaintext FL ($d\approx10^5$)	$\sim$4.3 MB	No crypto

a) Layering.: Table I summarizes the division of labor: ML-KEM for PQ transit, ZKP+binding+Enc-consistency for oversized and split attacks, threshold open for curious aggregators, median for in-bound Byzantine. None of the layers alone meets the joint medical threat model; the contribution is their composition with explicit binding.

b) Deployment.: In a hospital consortium, each site runs a client; a neutral coordinator verifies proofs; t-of-n share holders (for example a DPO and two hospital CISOs) jointly open per-client ciphertexts; median aggregation yields the round update. Daily or weekly training absorbs the observed $\sim$20$\times$ CPU overhead vs. plaintext FedAvg (Fig. 5). For imaging-scale d, HE chunks and Unruh proofs grow with d; tabular/compact models (UCI) validate the crypto pipeline first.

VII. Limitations

Bit-level Keccak-f[1600] carries algebraic lane lemmas (θ-column parity, ρ rotations, π bijection, χ local form, ι locality, packing invertibility, 24-round fold) in EasyCrypt with a FIPS 202 executable checker. Full-vector HE+Unruh cost scales with d, but a shared SIS commitment matrix and modular BFV multiplication keep `ConvNet28`-scale runs interactive on CPU.

VIII. Conclusion

ZKFL-PQ combines ML-KEM-768, ciphertext-bound Unruh proofs with Enc-consistency, threshold BFV partial decryption, and default median aggregation for quantum-resilient medical FL. Experiments on synthetic, tabular, and imaging tasks show that ℓ_2 proofs alone are insufficient against in-bound attacks, while the full stack remains practical on CPU. Artifact: https://github.com/edlansiaux/pq-zkfl-medical.

Acknowledgment

STaR-AI (CHU de Lille) and M2 MIAS (Université de Lille).

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